Experiment 2(B): Optimum Frequency Identification

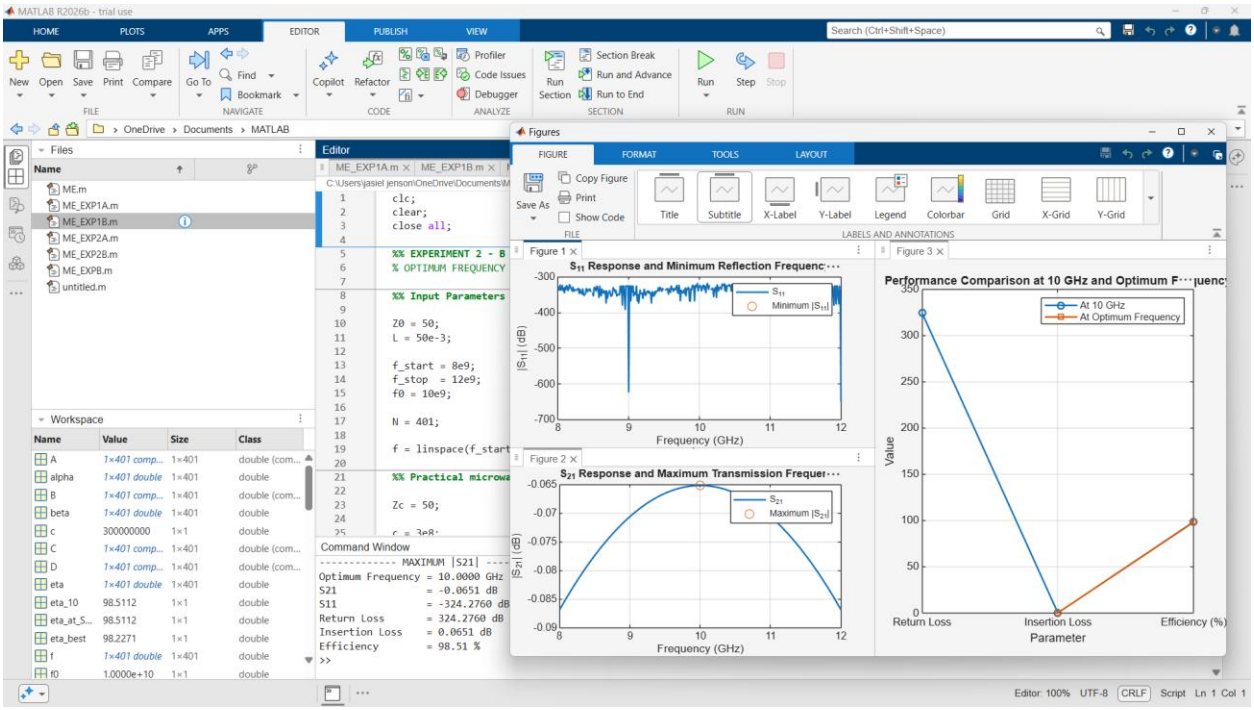
Aim:

To identify the optimum operating frequency by finding the minimum reflection coefficient and maximum transmission coefficient within the 8–12 GHz range.

Procedure:

1. Obtain the S-parameter response over 8–12 GHz.
2. Identify the frequency where \(|S_{11}|\) is minimum and \(|S_{21}|\) is maximum.
3. Calculate return loss and insertion loss at this frequency.
4. Compare the results with those at 10 GHz.

Output:



Result:

The optimum operating frequency was identified from the S-parameter sweep, and its matching and transmission performance were compared with those at 10 GHz.